DISHA BHARTI COLLEGE OF MANAGEMENT AND EDUCATION, SRE

FIRST SESSIONAL EXAMINATION - 2026

Roll No: _____

Class: BCA - Semester V
Subject: Java Programming

Time: 2 hours Maximum Marks: 60

General Instructions: Attempt all questions.

Q1. Attempt all parts - Very Short Answer ((5x2=10))

a) What is JVM? Explain its role.
b) Difference between JDK and JRE.
c) What is a Constructor? Give its types.
d) Define OOPs and its 4 pillars.
e) What is this keyword in Java?

Q2. Short Answer Type - Attempt any 5 ((5x3=15))

a) Explain method overloading with example.
b) What is Inheritance? Explain its types in Java.
c) Difference between String, StringBuffer and StringBuilder.
d) Explain try-catch-finally with program.
e) What is interface? How it is different from abstract class?

Q3. Explain in detail ((5x2=10))

a) Explain Package. How to create and import a package?
b) What is Exception Handling? Explain checked vs unchecked exception.
c) Explain Multithreading and its life cycle.
d) What is File Handling? Write a program to read/write a file.
e) Explain Access Modifiers in Java (public, private, protected).

Q4. Long Answer - Code Based ((5x3=15))

a) Write a Java program to implement Single Inheritance.
b) Write a program to demonstrate Method Overriding.
c) Write a program to create a Thread by extending Thread class and Runnable interface.
d) Write a program to demonstrate Polymorphism.
e) Write a Java program to find factorial using Recursion.

Q5. Very Long Answer - Attempt any 2 ((2x5=10))

a) What is AWT? Explain its components and write a program to create a simple GUI.
b) Explain JDBC. Write steps to connect Java with MySQL database.
c) Explain Collection Framework in detail with its hierarchy.

Q6. Objective / MCQ Type ((5x2=10))

a) Which keyword is used to inherit a class? (extends)
b) Java is platform independent because of?
c) Which method is entry point of Java program? (main)
d) Size of int in Java?
e) Can we overload main method? Yes/No with reason.